Data Communications & Networks

Chapter 3: Sections 3.3, 3.4, 3.5 & 3.6

Digital Signals, Transmission Impairment, Data Rate Limits & Network Performance

Topics Covered:

- Section 3.3 Digital Signals: levels, bit rate, bit length, baseband vs broadband transmission
- Section 3.4 Transmission Impairment: Attenuation (dB), Distortion, Noise (types), SNR
- Section 3.5 Data Rate Limits: Nyquist & Shannon formulas
- Section 3.6 Network Performance: Bandwidth, Throughput, Latency, Bandwidth-Delay Product, Jitter

Reference: Forouzan, *Data Communications and Networking*, 5th ed.

April 14, 2026

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Chapter 1 Digital Signals — Section 3.3

Analog vs Digital Signals: The Core Difference

So far in Chapter 3, Forouzan has been discussing **analog** signals — smooth continuous waves. But computers don't work with smooth waves. Computers work with **discrete** values: 0s and 1s.

A **digital signal** is one that can only take a *limited set of values*. The simplest case has two values (two levels): a high voltage for 1 and a low (or zero) voltage for 0. This is exactly how a CPU, RAM, and hard drives communicate internally.

The key question of Section 3.3 is: *What are the properties of digital signals and how do we transmit them on real-world physical links?*

1.1 Signal Levels and Bits per Level

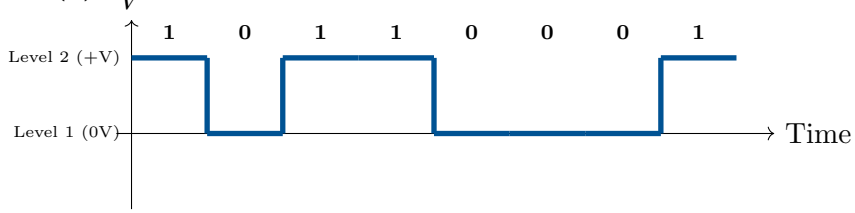
Why Use More Than Two Levels?

A digital signal doesn't have to use just two levels (0V and +V). We can use **multiple voltage levels**, where each unique voltage pattern encodes *multiple bits* at once.

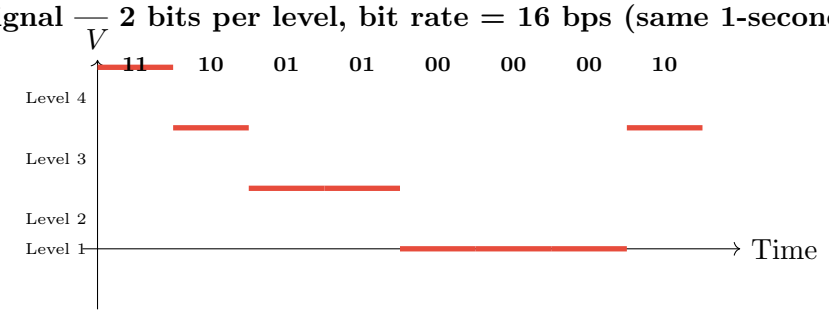
General rule: If a signal has L discrete levels, each level encodes $\log_2 L$ bits.

Levels (L)	Bits per level	Example voltages	Use case
2	$\log_2 2 = 1$	0V, +5V	Simple binary
4	$\log_2 4 = 2$	0V, +1V, +2V, +3V	2B1Q line coding
8	$\log_2 8 = 3$	0,1,2,3,4,5,6,7 V	Some DSL schemes
16	$\log_2 16 = 4$	16 distinct voltage steps	Advanced modems

(a) 2-level signal — 1 bit per level, bit rate = 8 bps



(b) 4-level signal — 2 bits per level, bit rate = 16 bps (same 1-second window)



Example 3.16 — 8-level digital signal

Question: A digital signal has 8 levels. How many bits are needed per level?

Solution:

$$\text{Bits per level} = \log_2 8 = \log_2 2^3 = \mathbf{3} \text{ bits/level}$$

Each time the voltage changes to a new level, 3 bits are conveyed simultaneously.

Example 3.17 — 9-level digital signal (non-power-of-2)

Question: A digital signal has 9 levels. How many bits per level?

Solution:

$$\log_2 9 \approx 3.17 \text{ bits/level}$$

This is *not* an integer! In practice, we **round up** to the next power of 2: use $L = 16$ (4 bits/level) or accept only $L = 8$ (3 bits/level) and not use the 9th level.

1.2 Bit Rate

Bit Rate — The Speed of a Digital Signal

Digital signals are **nonperiodic** (they don't repeat a pattern). So we cannot use "frequency" to describe how fast they change. Instead, we use **bit rate**: the number of bits transmitted per second.

$$\text{Bit rate [bps]} = \frac{\text{total bits sent}}{\text{time (seconds)}}$$

Common units: bps, kbps (10^3), Mbps (10^6), Gbps (10^9), Tbps (10^{12}).

Example 3.18 — Downloading text at 100 pages/second

Given: Download text at 100 pages/s. Each page = 24 lines × 80 characters; each character = 8 bits. Find required bit rate.

Solution:

$$\text{Bit rate} = 100 \times 24 \times 80 \times 8 = 1{,}536{,}000 \text{ bps} = \mathbf{1.536} \text{ Mbps}$$

Example 3.19 — Digitised voice channel

Given: A voice signal has 4 kHz bandwidth. To digitise it we sample at twice the highest frequency (Nyquist sampling theorem). Each sample = 8 bits.

Solution:

$$\text{Samples/second} = 2 \times 4000 = 8000 \text{ samples/s}$$

$$\text{Bit rate} = 8000 \times 8 = \mathbf{64{,}000} \text{ bps} = \mathbf{64} \text{ kbps}$$

This is the origin of the famous DS-0 (T-carrier) channel rate of 64 kbps.

Example 3.20 — High-definition TV (HDTV) bit rate

Given: HDTV resolution: 1920 × 1080 pixels; 30 frames/s; 24 bits/pixel (colour).

Solution:

$$\text{Bit rate} = 1920 \times 1080 \times 30 \times 24 = \mathbf{1{,}492{,}992{,}000} \approx \mathbf{1.5} \text{ Gbps}$$

TV stations compress this to 20–40 Mbps (using codecs like H.264/HEVC). The raw 1.5 Gbps shows why video compression is critical!

1.3 Bit Length

Bit Length

Analogous to wavelength for analog signals, **bit length** is the physical distance that one bit occupies on the transmission medium:

$$\text{Bit length} = \text{propagation speed} \times \text{bit duration} = \text{propagation speed} \times \frac{1}{\text{bit rate}}$$

For example, on a cable with propagation speed 2×10^8 m/s and bit rate 100 Mbps:

$$\text{Bit length} = \frac{2 \times 10^8}{10^8} = \mathbf{2} \text{ m/bit}$$

Each bit "occupies" 2 metres of cable!

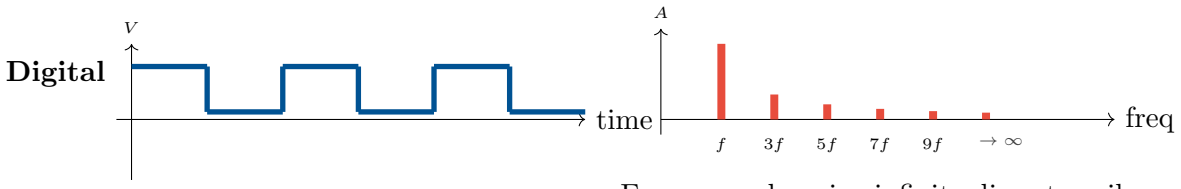
1.4 Digital Signal as a Composite Analog Signal

Fourier Analysis of Digital Signals

By Fourier analysis, *any* signal — including a digital square wave — can be decomposed into a sum of sine waves.
A digital signal has **instantaneous vertical transitions** (infinite rate of change) and **flat horizontal segments** (zero rate of change). Representing these extremes requires frequencies from **0 to infinity**.
Conclusion: A digital signal has **infinite bandwidth**.

Signal type	Frequency content	Bandwidth
Periodic digital	Discrete: $f, 3f, 5f, 7f, \ldots$	Infinite (discrete)
Nonperiodic digital	Continuous: 0 to ∞	Infinite (continuous)

This is a big problem: real channels have *finite* bandwidth. We can't send a perfect digital signal on a real cable! But we can send *enough harmonics* to make the signal recognisable.

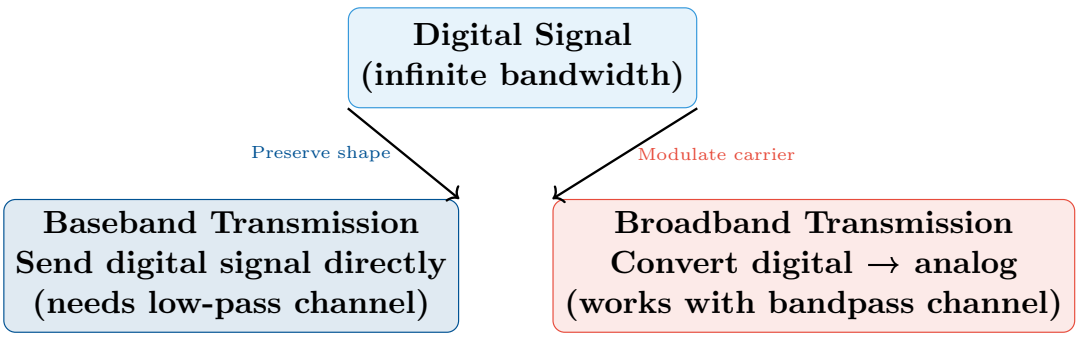


Frequency domain: infinite discrete spikes
(odd harmonics only for square wave)

1.5 Transmission of Digital Signals

Two Approaches: Baseband vs Broadband

Since a digital signal has infinite bandwidth but real channels have finite bandwidth, how do we actually transmit digital data? Two approaches:



1.5.1 Baseband Transmission

Baseband: Send the Digital Signal As-Is

Baseband transmission sends the digital signal directly without converting it to analog. This requires a **low-pass channel** — one whose bandwidth starts from zero (DC).

- **Wide low-pass channel**: Passes almost all harmonics of the digital signal. Received signal looks nearly identical to sent signal. Example: dedicated Ethernet cable between two computers.
- **Narrow low-pass channel**: Only a few harmonics pass through. The received signal is a rough approximation. We must decide: how many harmonics are enough to recognise the bits?

Required Bandwidth for Baseband Transmission

The **worst case** for a digital signal is alternating 1s and 0s (`010101...`). This needs the highest frequency component $f = N/2$ (where N is the bit rate).

Baseband Bandwidth Requirements

Harmonics used	Required bandwidth	Approximation quality
1st harmonic only	$B = \frac{N}{2}$	Rough (staircase shape)
1st + 3rd harmonics	$B = \frac{3N}{2}$	Better
1st + 3rd + 5th harmonics	$B = \frac{5N}{2}$	Good
∞ harmonics	$B = \infty$	Perfect (not possible)

Bit rate	1st harmonic	1st + 3rd	1st + 3rd + 5th
1 kbps	500 Hz	1.5 kHz	2.5 kHz
10 kbps	5 kHz	15 kHz	25 kHz
100 kbps	50 kHz	150 kHz	250 kHz

Example 3.22 — Bandwidth for 1 Mbps baseband

Question: What bandwidth is needed to send 1 Mbps via baseband?

Solution:

1. **Rough** (1st harmonic only): $B = \frac{N}{2} = \frac{10^6}{2} = \mathbf{500}$ kHz
2. **Better** (1st + 3rd harmonic): $B = 3 \times 500 = \mathbf{1.5}$ MHz
3. **Good** (1st + 3rd + 5th harmonic): $B = 5 \times 500 = \mathbf{2.5}$ MHz

Example 3.23 — Max bit rate for 100-kHz low-pass channel

Question: A low-pass channel has 100 kHz bandwidth. What is the maximum bit rate using the first harmonic approximation?

Solution: With 1st harmonic: $B = N/2 \Rightarrow N = 2B$

$$N = 2 \times 100{,}000 = \mathbf{200} \text{ kbps}$$

1.5.2 Broadband Transmission (Modulation)

Broadband: Convert Digital to Analog First

A **bandpass channel** (like a telephone line, cable TV channel, or wireless radio band) cannot pass DC (zero frequency). It has a range like 300–3400 Hz or 88–108 MHz.
The solution: **modulate** the digital signal onto an analog carrier wave. The carrier's frequency is chosen to sit within the available bandpass channel.

- This is exactly what a **modem** does (MOdulator-DEModulator).
- At the sender: digital → analog.
- At the receiver: analog → digital.

We study the specific modulation methods (ASK, FSK, PSK, QAM) in Chapter 5.

Example 3.24 — Telephone modem (broadband over voice line)

The telephone subscriber line carries 0–4 kHz. Treated as low-pass with 4 kHz bandwidth:

$$N_{\text{max}} = 2 \times 4000 = 8000 \text{ bps} \quad \text{(only 8 kbps!)}$$

A 56K modem treats the same line as a **bandpass** channel and uses sophisticated modulation (QAM) to achieve 56 kbps — 7 times more! This is the power of broadband transmission.

Example 3.25 — Digital cellular phones

A cellular network allocates a slice of its wide spectrum to each call. Even though the total network bandwidth is very wide, each user only gets a narrow bandpass slice (e.g., W/1000 for 1000 simultaneous calls). The phone converts voice to digital, then converts digital to an analog modulated signal that fits in its allocated bandpass slice. This is broadband transmission.

Chapter 2 Transmission Impairment — Section 3.4

Why Signals Degrade

In a perfect world, what you send is exactly what arrives. In the real world, **transmission media are not perfect**. The signal suffers from three types of impairment as it travels:

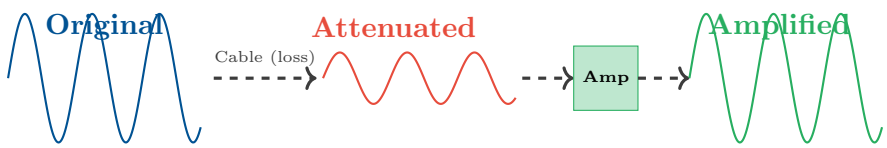
Three Causes of Signal Impairment

Attenuation (Energy loss)	Distortion (Shape change)	Noise (Added garbage)

2.1 Attenuation — 3.4.1

Attenuation: The Signal Loses Energy

When a signal travels through a medium (copper wire, fibre, air), it loses energy to the **resistance** of the medium. The signal becomes weaker (lower amplitude) over distance. This loss is called **attenuation**.
Real-world consequence: If the signal becomes too weak, the receiver can't distinguish 1s from 0s. That's why long-distance cables need **amplifiers** or **repeaters** at regular intervals to boost the signal back up.



Attenuation: signal weakens over distance; amplifiers restore strength

2.1.1 Decibels (dB): Measuring Attenuation

Decibel Formula

Engineers measure signal strength changes in **decibels (dB)**:

$$\text{dB} = 10\ \log_{10}\left(\frac{P_2}{P_1}\right)$$

- P_1 = signal power at point 1 (watts or milliwatts)
- P_2 = signal power at point 2
- **Negative dB** = attenuation (power decreased)
- **Positive dB** = amplification (power increased)

Key shortcut: Decibels *add* across multiple links: $\text{dB}_{\text{total}} = \text{dB}_1 + \text{dB}_2 + \text{dB}_3 + \cdots$ This makes cascaded networks easy to analyse without multiplication.

Memory shortcuts for dB:

- Power halved $\Rightarrow -3$ dB (loss of 3 dB)
- Power doubled $\Rightarrow +3$ dB
- Power $\times 10 \Rightarrow +10$ dB
- Power $\div 10 \Rightarrow -10$ dB

Example 3.26 — Signal power reduced to one-half

Given: $P_2 = \frac{1}{2}P_1$ (half the power lost in cable).

Solution:

$$\text{dB} = 10\log_{10}\left(\frac{0.5\,P_1}{P_1}\right) = 10\log_{10}(0.5) = 10 \times (-0.301) = \mathbf{-3}\ \text{dB}$$

A loss of half the power = -3 dB. This is the most important dB relationship to memorise!

Example 3.27 — Amplifier increases power by 10×

Given: $P_2 = 10\,P_1$ (amplifier boosts signal 10 times).

Solution:

$$\text{dB} = 10\log_{10}(10) = 10 \times 1 = +\mathbf{10}\ \text{dB}$$

An amplifier with +10 dB gain multiplies power by 10.

Example 3.28 — Cascaded links: add dBs

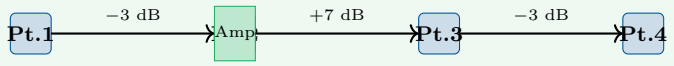
Given: Signal travels through 4 points:

- Point 1 $\rightarrow$ 2: -3 dB (cable attenuation)
- Point 2 $\rightarrow$ 3: $+7$ dB (amplifier)

- Point 3 → 4: −3 dB (cable attenuation)

Solution: Just add the dBs:

$$\text{Total} = -3 + 7 + (-3) = +\mathbf{1}\ \text{dB}$$



Net gain = −3 + 7 − 3 = +1 dB (signal slightly stronger at end)

Example 3.29 — dB_m: power in milliwatts

Given: $\text{dB}_m = -30$. Find power in milliwatts.

Note: dB_m measures absolute power referenced to 1 mW: $\text{dB}_m = 10\log_{10}(P_m)$ where P_m is in milliwatts.

Solution:

$$-30 = 10\log_{10}(P_m) \;\Rightarrow\; \log_{10}(P_m) = -3 \;\Rightarrow\; P_m = 10^{-3} = \mathbf{0.001}\ \text{mW} = \mathbf{1}\ \mu\text{W}$$

Example 3.30 — Cable loss in dB/km (practical!)

Given: Cable loss = −0.3 dB/km. Signal power at start = 2 mW. Find power after 5 km.

Solution:

$$\text{Total dB loss} = 5 \times (-0.3) = -1.5\ \text{dB}$$

$$-1.5 = 10\log_{10}\left(\frac{P_2}{2}\right) \;\Rightarrow\; \log_{10}\left(\frac{P_2}{2}\right) = -0.15 \;\Rightarrow\; \frac{P_2}{2} = 10^{-0.15} = 0.708$$

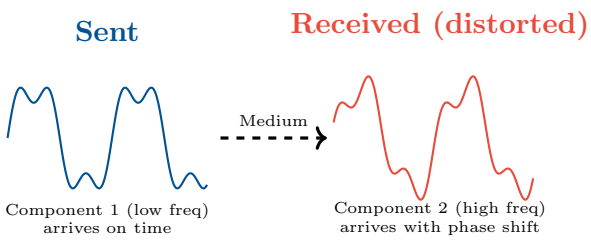
$$P_2 = 0.708 \times 2 = \mathbf{1.416}\ \text{mW}$$

About 30% of the signal power is lost over 5 km of cable.

2.2 Distortion — 3.4.2

Distortion: The Signal Changes Shape

A composite signal is made of many frequency components. Each component travels at a slightly different speed through the medium (different **propagation speed** for different frequencies). They arrive at different times, causing a **phase shift**. The result: the received composite signal has a different shape from the original. This is **distortion**.
Example: In a digital signal, the high-frequency components (sharp edges) arrive slightly later than the low-frequency components. The sharp square edges become rounded and slurred — making it harder to tell where one bit ends and the next begins.



Distortion: components arrive at different times → different shape

2.3 Noise — 3.4.3

Noise: Unwanted Signals Added to Yours

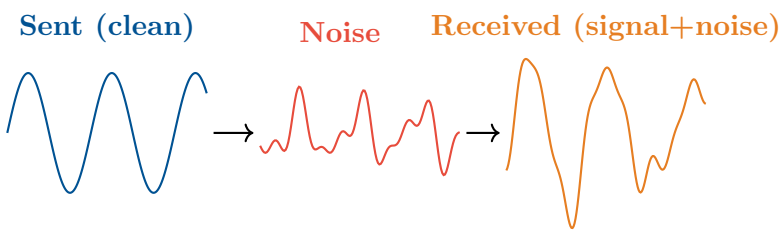
Noise is any unwanted electrical energy that gets mixed into the signal. Unlike attenuation (which just weakens your signal) and distortion (which changes its shape), noise *adds a completely foreign signal* on top of yours.
The received signal = original signal + noise.
Four types of noise:

1. **Thermal noise**: Caused by random motion of electrons in any conductor due to heat. Unavoidable — even a perfectly shielded wire has thermal noise. Measured as kTB (Boltzmann's constant × temperature × bandwidth).

2. **Induced noise**: From external sources like motors, fluorescent lights, and household appliances. They act as antennas and transmit electromagnetic interference into nearby cables.

3. **Crosstalk**: One wire "leaks" its signal into a neighbouring parallel wire (the way radio stations leak into adjacent frequencies). Common in telephone cables with many wires bundled together.

4. **Impulse noise**: A sudden spike of very high energy lasting a very short time. Caused by lightning strikes, power surges, or switching equipment. Can destroy many bits at once — the hardest noise type to handle.



Noise adds a random "fuzz" on top of the real signal

2.3.1 Signal-to-Noise Ratio (SNR)

SNR — Measuring How Clean the Channel Is

$$\text{SNR} = \frac{\text{Average Signal Power}}{\text{Average Noise Power}}$$

In decibels:

$$\text{SNR}_{dB} = 10\ \log_{10}(\text{SNR})$$

- **High SNR** (e.g., 10 000 = 40 dB): signal is much stronger than noise. Channel is clean; high data rates are achievable.
- **Low SNR** (e.g., 10 = 10 dB): noise is significant. Channel is noisy; data rates are limited.
- **SNR = ∞**: perfectly noiseless (theoretical ideal only).

Example 3.31 — Computing SNR and SNR_{dB}

Given: Signal power = 10 mW; noise power = 1 μW = 0.001 mW.

Solution:

$$\text{SNR} = \frac{10\ \text{mW}}{0.001\ \text{mW}} = \mathbf{10{,}000}$$

$$\text{SNR}_{dB} = 10\log_{10}(10{,}000) = 10\log_{10}(10^4) = 10 \times 4 = \mathbf{40}\ \text{dB}$$

The signal is 10 000 times more powerful than the noise. Very clean channel!

Example 3.32 — Ideal noiseless channel

Given: No noise at all (noise power = 0).

Solution:

$$\text{SNR} = \frac{P_{\text{signal}}}{0} = \infty \qquad \Rightarrow \qquad \text{SNR}_{dB} = 10\log_{10}(\infty) = \infty$$

This is purely theoretical. Every real channel has some thermal noise.

Chapter 3 Data Rate Limits — Section 3.5

Why Data Rate Limits Matter

Given a channel with a certain bandwidth and noise level, what is the absolute *maximum* number of bits per second we can send? Two mathematicians answered this:

- **Harry Nyquist** (1928): formula for a *noiseless* channel.
- **Claude Shannon** (1948): formula for a *noisy* channel.

These are theoretical *upper bounds*. No engineering trick can exceed them. They define what is physically possible.

3.1 Nyquist Bit Rate — Noiseless Channel

Nyquist Bit Rate Formula

For a **noiseless** channel:

$$\text{BitRate} = 2 \times \text{Bandwidth} \times \log_2 L$$

- **Bandwidth** = channel bandwidth in Hz
- **L** = number of signal levels
- $\log_2 L$ = bits per signal level
- Factor of 2 comes from Nyquist's proof that the maximum baud rate is $2B$

Example 3.33 — Nyquist agrees with baseband intuition

Recall from Section 3.3: baseband transmission with 1st harmonic gives $N = 2B$ (for 2 levels). Nyquist confirms: $2 \times B \times \log_2 2 = 2B$. They match! The Nyquist formula is a generalisation for any number of levels.

Example 3.34 — 3000-Hz channel, 2 levels

$$\text{BitRate} = 2 \times 3000 \times \log_2 2 = 2 \times 3000 \times 1 = \mathbf{6000} \text{ bps}$$

Example 3.35 — 3000-Hz channel, 4 levels

$$\text{BitRate} = 2 \times 3000 \times \log_2 4 = 2 \times 3000 \times 2 = \mathbf{12{,}000} \text{ bps}$$

Doubling levels doubled the bit rate!

Example 3.36 — Find L for 265 kbps on 20-kHz channel

Given: Need 265 kbps on a 20-kHz noiseless channel. How many signal levels?

Solution:

$$265{,}000 = 2 \times 20{,}000 \times \log_2 L \;\Rightarrow\; \log_2 L = \frac{265{,}000}{40{,}000} = 6.625 \;\Rightarrow\; L = 2^{6.625} \approx 98.7$$

Not a power of 2. Practical options:

- Use $L = 128$: BitRate $= 2 \times 20{,}000 \times 7 = 280$ kbps (slightly above target)
- Use $L = 64$: BitRate $= 2 \times 20{,}000 \times 6 = 240$ kbps (slightly below target)

3.2 Shannon Capacity — Noisy Channel

Shannon Channel Capacity Formula

For a **noisy** channel, the absolute theoretical maximum (the "capacity") is:

$$C = B \times \log_2(1 + \text{SNR})$$

- C = channel capacity (bps)
- B = bandwidth (Hz)
- SNR = signal-to-noise ratio (plain ratio, not in dB!)
- Shannon showed: no matter how clever the encoding, you *cannot* exceed C.
- The formula gives the ceiling; it doesn't tell you how to achieve it.

Example 3.37 — SNR ≈ 0 (extreme noise)

$$C = B \log_2(1 + 0) = B \log_2(1) = B \times 0 = \mathbf{0} \text{ bps}$$

If noise drowns the signal completely, zero data can be transferred, regardless of bandwidth.

Example 3.38 — Telephone line capacity

Given: Telephone line bandwidth = 3000 Hz; SNR = 3162.

Solution:

$$C = 3000 \times \log_2(1 + 3162) = 3000 \times \log_2(3163)$$

$$\log_2(3163) = \frac{\ln 3163}{\ln 2} = \frac{8.059}{0.693} \approx 11.63$$

$$C = 3000 \times 11.63 \approx \mathbf{34{,}860} \text{ bps} \approx \mathbf{35} \text{ kbps}$$

This is why old dial-up modems maxed out at 33.6 kbps — Shannon's law imposed the ceiling.

Example 3.39 — SNR given in dB

Given: $\text{SNR}_{dB} = 36$ dB; bandwidth = 2 MHz. Find capacity.

Step 1: Convert SNR_{dB} to plain SNR:

$$\text{SNR} = 10^{\text{SNR}_{dB}/10} = 10^{36/10} = 10^{3.6} = 3981$$

Step 2: Apply Shannon:

$$C = 2 \times 10^6 \times \log_2(1 + 3981) = 2 \times 10^6 \times \log_2(3982)$$

$$\log_2(3982) \approx 11.96$$

$$C \approx 2 \times 10^6 \times 12 = \mathbf{24} \text{ Mbps}$$

Example 3.40 — High-SNR shortcut

When SNR is very large: $\log_2(1 + \text{SNR}) \approx \log_2(\text{SNR}) \approx \text{SNR}_{dB}/3$.
For Example 3.39: $C \approx 2$ MHz $\times (36/3) = 2 \times 12 = \mathbf{24}$ Mbps. Same answer, much simpler calculation!

3.3 Using Both Limits Together — 3.5.3

The Correct Workflow: Shannon then Nyquist

1. **Use Shannon** to find the absolute capacity ceiling C. This is the maximum bit rate possible on this channel regardless of encoding.
2. **Choose a target bit rate** below the Shannon limit (for reliability margin).
3. **Use Nyquist** to find the number of signal levels L needed to achieve that target bit rate on the channel bandwidth.

This two-step approach is what real engineers and modem designers use.

Example 3.41 — Design a 1-MHz channel with SNR = 63

Given: Channel bandwidth = 1 MHz; SNR = 63.

Step 1 — Shannon (ceiling):

$$C = 10^6 \times \log_2(1 + 63) = 10^6 \times \log_2(64) = 10^6 \times 6 = \mathbf{6} \text{ Mbps}$$

Step 2 — Choose target: Pick 4 Mbps (below the 6 Mbps ceiling for safety

margin).

Step 3 — Nyquist (find L):

$$4{,}000{,}000 = 2 \times 1{,}000{,}000 \times \log_2 L \;\Rightarrow\; \log_2 L = 2 \;\Rightarrow\; L = 2^2 = \mathbf{4} \text{ signal levels}$$

Conclusion: Use 4-level signalling to achieve 4 Mbps on this channel. Shannon tells you what's possible; Nyquist tells you how many levels to use.

Two-step design process

Chapter 4 Network Performance — Section 3.6

Five Performance Metrics

After understanding *what* we can send (data rate limits), we ask: *how well* does the network perform in practice? Section 3.6 introduces five key metrics:

Metric	Unit	Simple meaning
Bandwidth	bps / Hz	Maximum theoretical capacity
Throughput	bps	Actual data rate achieved
Latency (Delay)	seconds	Time from send to receive
BW-Delay Product	bits	Max bits "in flight" on the link
Jitter	seconds	Variation in packet delays

4.1 Bandwidth

Two Meanings of Bandwidth

- **Bandwidth in Hz**: The frequency range a medium can carry. Example: phone line = 4 kHz; fibre = hundreds of THz.
- **Bandwidth in bps**: The maximum data rate on a link. Example: "we have a 100 Mbps link."

More Hz ⇒ higher possible bps (via Nyquist/Shannon). They are related but not the same thing.

Example 3.41 (Section 3.6) — Subscriber line

A telephone subscriber line has 4 kHz bandwidth for voice. For data, a sophisticated modem can push up to 56 kbps through the same 4 kHz. The bandwidth in Hz didn't change; clever modulation (more signal levels) raised the bandwidth in bps.

4.2 Throughput

Throughput: The Real-World Speed

Throughput is what you actually measure — not the theoretical limit.

$$\text{Throughput} \leq \text{Bandwidth}$$

Throughput is lower than bandwidth because of:

- Network congestion (shared links)
- Protocol overheads (headers, ACKs, retransmissions)
- Hardware bottlenecks
- Errors causing retransmissions

Example: You have a 1 Gbps Ethernet card but download a file at 150 Mbps. Your throughput = 150 Mbps; your bandwidth = 1 Gbps.

4.3 Latency (Delay)

Total Latency Formula

$$\text{Latency} = \underbrace{\frac{d}{s}}_{\text{propagation}} + \underbrace{\frac{n}{B}}_{\text{transmission}} + t_{\text{queue}} + t_{\text{processing}}$$

- d = distance (m); s = propagation speed (m/s); *propagation time* = physical travel time of the signal
- n = message size (bits); B = bandwidth (bps); *transmission time* = time to put all bits on the wire
- t_{queue} = waiting time in router buffer
- $t_{\text{processing}}$ = time router spends looking up next hop

Example 3.45 — Propagation time across 12,000 km

Given: Distance = 12,000 km; propagation speed = 2.4×10^8 m/s.

$$t_{\text{prop}} = \frac{12{,}000{,}000}{2.4 \times 10^8} = \frac{1.2 \times 10^7}{2.4 \times 10^8} = 0.05 \text{ s} = \mathbf{50} \text{ ms}$$

Example 3.46 — Transmission dominates (large message, slow link)

Given: Message = 100 MB = 800×10^6 bits; bandwidth = 1 Mbps; distance = 1000 km; speed = 2×10^8 m/s.

$$t_{\text{trans}} = \frac{800 \times 10^6}{10^6} = \mathbf{800}\ \text{s} \qquad t_{\text{prop}} = \frac{10^6}{2 \times 10^8} = 0.005\ \text{s} = \mathbf{5}\ \text{ms}$$

Total $\approx$ 800 s. Propagation is negligible; **transmission dominates**.

Example 3.47 — Propagation dominates (single bit, fast link)

Given: Message = 1 bit; bandwidth = 1 Gbps; distance = 1000 km.

$$t_{\text{trans}} = \frac{1}{10^9} = 1\ \text{ns} \qquad t_{\text{prop}} = \frac{10^6}{2 \times 10^8} = 5\ \text{ms}$$

Total $\approx$ 5 ms. Transmission is negligible; **propagation dominates**.

Example P3.33 — Complete latency calculation

Given:

- Frame size: 5 000 000 bits
- Link length: 2000 km; propagation speed = 2×10^8 m/s
- Bandwidth: 10 Mbps
- 10 routers, each: queuing time = 2 μs, processing time = 1 μs

Solution:

$$\begin{aligned}
t_{\text{trans}} &= \frac{5{,}000{,}000}{10{,}000{,}000} = 0.5\ \text{s} \\
t_{\text{prop}} &= \frac{2{,}000{,}000}{2 \times 10^8} = 0.01\ \text{s} = 10\ \text{ms} \\
t_{\text{queue}} &= 10 \times 2 \times 10^{-6} = 20\ \mu\text{s} \\
t_{\text{proc}} &= 10 \times 1 \times 10^{-6} = 10\ \mu\text{s}
\end{aligned}$$

$$\text{Total} = 0.5 + 0.01 + 0.00002 + 0.00001 \approx \mathbf{0.510}\ \text{s} \approx \mathbf{510}\ \text{ms}$$

Transmission time (500 ms) completely dominates; routing overhead is negligible.

4.4 Bandwidth-Delay Product

Bandwidth-Delay Product

$$\text{BDP} = \text{Bandwidth} \times \text{Propagation delay}$$

Tells you the maximum number of bits that can be "in flight" on the link at any given moment — the "volume" of the pipe.
Pipe analogy: Bandwidth = diameter; propagation delay = length; BDP = volume.

Example 3.49 — Narrow/short pipe (BDP = 5 bits)

Given: Bandwidth = 1 bps; propagation delay = 5 s.

$$\text{BDP} = 1 \times 5 = \mathbf{5} \text{ bits}$$

At any moment, at most 5 bits can be on the wire simultaneously.

Example 3.50 — Wide pipe (BDP = 25 bits)

Given: Bandwidth = 5 bps; propagation delay = 5 s.

$$\text{BDP} = 5 \times 5 = \mathbf{25} \text{ bits}$$

Real-world BDP — Trans-Atlantic 1 Gbps fibre

Given: Bandwidth = 1 Gbps; propagation delay ≈ 50 ms.

$$\text{BDP} = 10^9 \times 0.05 = 50{,}000{,}000 \text{ bits} = \mathbf{50} \text{ Mb} = \mathbf{6.25} \text{ MB}$$

Over 6 MB of data can be "in flight" at once across the Atlantic! This is why TCP needs large receiver window sizes for efficient use of high-BDP links.

4.5 Jitter

Jitter: Inconsistent Delays

Jitter = variation in the arrival time of successive packets.

- If packets are sent every 20 ms, the receiver expects them every 20 ms.
- Due to varying queue lengths and different paths, actual arrivals might be: 18 ms, 25 ms, 19 ms, 28 ms, ...
- This variation is jitter.
- **Critical for real-time traffic:** VoIP, video calls, online games. High jitter ⇒ choppy audio, frozen video, lag spikes.
- **Solution:** Playout buffer. Packets arrive early and wait in the buffer; they are played out at a fixed, regular rate. The buffer absorbs the jitter. Too small a buffer ⇒ gaps. Too large ⇒ adds delay.

4.6 Master Summary: All Metrics

Metric	Formula	Unit	Dominant when...
Bandwidth (Hz)	—	Hz	—
Bandwidth (bps)	—	bps	—
Throughput	Measured	bps	Always $\leq$ BW
Propagation time	d/s	s	Long distance, small msg
Transmission time	n/B	s	Large msg, slow link
Total Latency	sum of 4 components	s	Depends on case
BW-Delay Product	$B \times t_{\text{prop}}$	bits	High-speed long links
Jitter	Variation in delay	s	Real-time applications

Complete Chapter Summary

Section 3.3 — Digital Signals

- L levels $\Rightarrow \log_2 L$ bits per level.
- Bit rate = bits/s; Bit length = prop. speed $\times$ bit duration.
- Digital signals have **infinite bandwidth** (Fourier).
- **Baseband**: send digital signal directly; needs low-pass channel; rough approx BW $= N/2$.
- **Broadband**: modulate onto carrier; works with bandpass channel; uses modem (ASK/FSK/PSK/QAM — covered in Ch. 5).

Section 3.4 — Transmission Impairment

- **Attenuation**: energy loss; measured in dB; dBs add over cascaded links. -3 dB = half the power; $+10$ dB = $10\times$ the power.
- **Distortion**: shape change; frequency components travel at different speeds.
- **Noise**: 4 types: thermal, induced, crosstalk, impulse. SNR = signal power / noise power. $\text{SNR}_{dB} = 10\log_{10}(\text{SNR})$.
- -3 dB = half power; -30 dBm = 10^{-3} mW = 1 μW.

Section 3.5 — Data Rate Limits

- **Nyquist** (noiseless): $\text{BitRate} = 2B\log_2 L$. More levels $\Rightarrow$ more bits/s, but less reliability.
- **Shannon** (noisy): $C = B\log_2(1 + \text{SNR})$. Absolute ceiling; no scheme can exceed it.
- **Workflow**: Shannon $\rightarrow$ ceiling; Nyquist $\rightarrow$ find L.

Section 3.6 — Network Performance

- **Bandwidth** (theoretical max) $\geq$ **Throughput** (actual).
- Latency = propagation (d/s) + transmission (n/B) + queuing + processing.
- Propagation dominates for small messages over long distances; transmission dominates for large messages over short, slow links.
- **BDP** $= B \times t_{\text{prop}}$ = bits in flight; important for protocol window sizing.
- **Jitter** = delay variation; critical for real-time apps; solved by playout buffer.